CUSTOMER SHOPPING BEHAVIOR ANALYSIS

An End-to-End Data Analytics Project Using Python, SQL and Power BI

Tools Used	Python (Pandas) · SQL / MySQL · Power BI
Dataset	Customer Shopping Behavior — 3,900 records · 18 columns
Report Type	Professional Portfolio Case Study
Analysis Date	August 2026

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1. EXECUTIVE SUMMARY

This report presents a comprehensive, end-to-end data analytics case study on customer shopping behavior. The analysis covers the full analytics lifecycle — data ingestion, cleaning, exploratory analysis, SQL querying, and interactive dashboard visualization — using Python (Pandas), MySQL, and Power BI.

Key Performance Indicators

3,900	$59.76	3.75
Number of Records	Average Purchase Amount	Average Review Rating

Figure 1.1 — Top-level KPIs (Power BI dashboard & Python df.describe())

Major Findings

- Clothing is the dominant category by both revenue (~$104K) and sales volume (~1,737 records).
- 73% of customers are non-subscribers; only 27% hold an active subscription.
- Young Adults (18–31) generate the highest revenue (~$62K) and lead in sales count (1,028).
- Discount Applied and Promo Code Used are 100% identical — a confirmed data quality issue.
- Shipping type has no material effect on purchase amount (~$59.76 across all types).
- The customer base is loyalty-mature: mean previous purchases = 25.35 (max 50).
- PayPal is the most-used payment method (677 records); 'Every 3 Months' is the top purchase cadence (584).

Tool	Purpose	Version / Notes
Python 3.12	Data loading, cleaning, feature engineering, MySQL export	pandas 3.0.0 · SQLAlchemy 2.0.51 · PyMySQL 1.2.0
MySQL	Structured querying, 10 business-question SQL analyses	Accessed via SQLAlchemy from Jupyter notebook
Power BI	Interactive dashboard — KPI cards, category & age charts	Customer Behavior Dashboard (.pbix)

Table 1.1 — Tools and Technologies

2. BUSINESS PROBLEM

Retail businesses generate large transaction volumes but often lack the infrastructure to convert raw records into actionable intelligence. Without clear insight into who is buying, what they buy, how frequently, and at what price, businesses cannot effectively target marketing spend, optimize product mix, or manage subscription revenue.

Business Questions Answered

#	Business Question
Q1	What is the total revenue generated by male vs. female customers?
Q2	Which customers used a discount but still spent above the average purchase amount?
Q3	Which products receive the highest average review ratings?
Q4	Does shipping type (Standard vs. Express) influence spend?
Q5	Do subscribed customers spend more than non-subscribers?
Q6	Which products have the highest discount application rates?
Q7	How is the customer base segmented by loyalty (New / Returning / Loyal)?
Q8	What are the top 3 products within each category by purchase volume?
Q9	Are repeat buyers (>5 purchases) more likely to be subscribers?

Q10	Which age group contributes the most revenue?

Table 2.1 — Ten business questions addressed in the SQL analysis

3. PROJECT OBJECTIVES

Analytical Objectives

- Load and inspect a 3,900-row retail dataset using Python / Pandas.
- Identify and treat missing values; remove redundant columns.
- Engineer age_group (quartile-based) and purchase_frequency_days (numeric) features.
- Export the cleaned dataset to MySQL for structured SQL analysis.
- Execute 10 targeted SQL queries covering demographics, product, and behavioral questions.
- Build an interactive Power BI dashboard with KPI cards, category charts, and slicers.

Business Objectives

- Identify highest-revenue categories and age segments to guide inventory and marketing decisions.
- Assess subscription program value by comparing subscriber vs. non-subscriber spend.
- Evaluate discount and promo usage relative to average purchase amounts.
- Segment customers into New, Returning, and Loyal groups for CRM targeting.
- Provide a self-service dashboard filterable by subscription, gender, category, and shipping type.

4. DATASET OVERVIEW

Dataset Profile

File	customer_shopping_behavior.csv
Rows	3,900 (confirmed: df.info() → RangeIndex 3900)
Original Columns	18
Final Columns	19 (18 − 1 dropped + 2 engineered)
Age Range	18 – 70 years (mean: 44.07)
Purchase Range	\$20 – \$100 (mean: \$59.76, median: \$60)
Categories	Clothing · Accessories · Footwear · Outerwear
Geographic Scope	50 US states

Data Dictionary

#	Column Name	Type	Description
1	Customer ID	int64	Unique transaction identifier (1–3,900)
2	Age	int64	Customer age, years (18–70, mean 44.07)
3	Gender	str	Male / Female (Male: 2,652; Female: 1,248)
4	Item Purchased	str	Product name — 25 unique items; most: Blouse (171)
5	Category	str	Clothing · Accessories · Footwear · Outerwear
6	Purchase Amount (USD)	int64	Transaction value \$20–\$100 (mean \$59.76, std \$23.69)
7	Location	str	US state — 50 unique; most frequent: Montana (96)
8	Size	str	S / M / L / XL (most common: M, 1,755 records)
9	Color	str	25 unique colors; most common: Olive (177)
10	Season	str	Spring / Summer / Fall / Winter; most: Spring (999)
11	Review Rating	float64	2.50–5.00 (mean 3.75, std 0.72) — 37 nulls imputed
12	Subscription Status	str	Yes (1,053 / 27%) or No (2,847 / 73%)

13	Shipping Type	str	6 types; most common: Free Shipping (675)
14	Discount Applied	str	Yes (1,677 / 43%) or No (2,223 / 57%)
15	Promo Code Used	str	100% identical to Discount Applied → dropped
16	Previous Purchases	int64	Count of prior purchases (1–50, mean 25.35)
17	Payment Method	str	6 types; most common: PayPal (677)
18	Frequency of Purchases	str	7 labels; most common: Every 3 Months (584)

Table 4.1 — Full data dictionary (18 original columns, verified from df.info() and df.describe())

Missing Values & Data Quality

df.isnull().sum() confirmed only one column with missing values: Review Rating — 37 nulls (0.95% of records). All other 17 columns had zero nulls. A critical quality finding: Discount Applied and Promo Code Used are perfectly correlated (100% identical across all 3,900 rows, confirmed programmatically). The redundant column was dropped, reducing the final clean dataset to 19 columns.

5. DATA CLEANING & PREPARATION USING PYTHON

5.1 Loading & Initial Inspection

```
import pandas as pd
df = pd.read_csv('../data/customer_shopping_behavior.csv')
df.info()          # 3,900 rows x 18 cols; Review Rating has 37 nulls
df.describe(include='all')   # summary statistics for all columns
```

Statistic	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
Count	3,900	3,900	3,863 (37 nulls)	3,900
Mean	44.07	$59.76	3.75	25.35
Std	15.21	$23.69	0.72	14.45
Min	18	$20	2.50	1
Median	44	$60	3.80	25
Max	70	$100	5.00	50

Table 5.1 — Descriptive statistics from df.describe() (notebook output)

5.2 Missing Value Treatment

```
# Impute Review Rating with category-level median (preserves category
distributions)
df['Review Rating'] = df.groupby('Category')['Review Rating'] \
    .transform(lambda x: x.fillna(x.median()))
df.isnull().sum()    # → all 0
```

After imputation, df.isnull().sum() confirmed 0 missing values across all columns.

5.3 Column Standardization

```
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

All column names converted to snake_case. Confirmed via df.columns output in the notebook.

5.4 Feature Engineering

Age Group — Quartile-Based Segmentation

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

Age Group	Approx. Age Range	Quartile
Young Adult	18 – 30	Q1 (bottom 25%)
Adult	31 – 44	Q2 (25th – 50th %)
Middle-aged	45 – 57	Q3 (50th – 75th %)
Senior	58 – 70	Q4 (top 25%)

Table 5.2 — Age group segmentation (pd.qcut with quartile boundaries Q1=31, Q2=44, Q3=57)

Purchase Frequency — Numeric Mapping

```
freq_map = {'Weekly':7, 'Fortnightly':14, 'Bi-Weekly':14,
            'Monthly':30, 'Quarterly':90, 'Every 3 Months':90, 'Annually':365}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(freq_map)
```

Converts the categorical frequency label into a numeric days-between-purchases value, enabling quantitative analysis (Weekly = 52 visits/year vs. Annual = 1 visit/year).

Redundant Column Removal

```
(df['discount_applied'] == df['promo_code_used']).all()  # → True
df = df.drop('promo_code_used', axis=1)  # reduces from 20 to 19 columns
```

5.5 Export to MySQL

```
from sqlalchemy import create_engine
engine =
create_engine('mysql+pymysql://root:root@localhost:3306/customer_behavior')
df.to_sql('customer', engine, if_exists='replace', index=False)
pd.read_sql('SELECT * FROM customer LIMIT 5;', engine)  # read-back confirmed
```

The 3,900-row × 19-column cleaned DataFrame was exported to the MySQL table customer in database customer_behavior, forming the source for all subsequent SQL queries.

6. EXPLORATORY DATA ANALYSIS

EDA findings are derived from the notebook's df.describe(include='all') output and the Power BI dashboard visuals. The notebook does not include matplotlib / seaborn chart cells; quantitative summaries are sourced directly from the verified notebook outputs.

6.1 Demographics

Dimension	Finding	Supporting Number
Gender	Male-dominant dataset	Male: 2,652 (68%) | Female: 1,248 (32%)
Age	Wide age spread, roughly uniform	Mean 44.07, Std 15.21, Range 18–70
Location	All 50 US states represented	Montana most frequent: 96 records
Season	Spring is the busiest season	Spring: 999 records (most frequent)

Table 6.1 — Key demographic findings (df.describe() output)

6.2 Category Performance

Category	Revenue (Dashboard)	Sales Count	Records (Python)
Clothing	~$104K (highest)	~1,600 (highest)	1,737 (df.describe top freq)
Accessories	~$75K	~1,100	Second most frequent
Footwear	~$38K	~600	Third tier
Outerwear	~$20K	~250	Lowest volume and revenue

Table 6.2 — Category performance (Power BI dashboard bar charts; Python count for Clothing confirmed at 1,737)

6.3 Purchase Amount & Ratings

- Mean purchase amount: $59.76 (confirmed in both Python describe() and Power BI KPI card).
- Median: $60.00 — near-identical to mean, indicating a symmetric distribution.
- Range: $20–$100 (integer values, suggesting rounded price tiers in source data).
- Mean review rating: 3.75 / 5.0 (Python: 3.750065; Power BI KPI: 3.75).
- Rating range: 2.50–5.00, Std: 0.72 — moderate clustering around the mean.

6.4 Subscription & Discount Behavior

Metric	Yes	No
Subscription Status	1,053 (27%) — Power BI donut confirmed	2,847 (73%) — df.describe freq=2847
Discount Applied	1,677 (43%)	2,223 (57%) — df.describe freq=2223

Table 6.3 — Subscription and discount distributions (Python & Power BI verified)

6.5 Shipping, Payment & Frequency

Dimension	Most Common	Count / Detail
Shipping Type	Free Shipping	675 records (df.describe freq=675); 6 types total
Payment Method	PayPal	677 records; 6 methods total
Frequency of Purchases	Every 3 Months	584 records; 7 frequency labels total
Previous Purchases	Mean: 25.35	Range 1–50, Std 14.45, Median 25

Table 6.4 — Shipping, payment, and frequency distributions

6.6 Age-Group Analysis

Age Group	Revenue (Dashboard)	Sales Count (Dashboard)	Rank
Young Adult	$62K	1,028	1st
Middle-aged	$59K	986	2nd
Senior	$56K	944	3rd
Adult	$56K	942	4th

Table 6.5 — Age-group revenue and sales (Power BI horizontal bar chart values)

7. SQL ANALYSIS

After cleaning, the 3,900-row DataFrame was exported to MySQL (database: customer_behavior, table: customer) via SQLAlchemy. Ten SQL queries addressed specific business questions. The most significant queries are presented below with code and business interpretation.

Q1 — Revenue by Gender

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer GROUP BY gender;
```

Male customers (68% of records) generate proportionally higher total revenue. Per-transaction spend is similar across genders given the uniform purchase-amount distribution.

Q2 — High-Value Discount Users

```
SELECT customer_id, purchase_amount FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Identifies customers who use discounts yet spend above the $59.76 average. This 'high-value discount user' cohort is better served by loyalty rewards than blanket discounts.

Q3 — Top 5 Products by Review Rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating), 2) AS `Average Product Rating`
FROM customer
GROUP BY item_purchased ORDER BY AVG(review_rating) DESC LIMIT 5;
```

Highest-rated products are candidates for featured placement and increased marketing emphasis.

Q4 — Shipping Type vs. Purchase Amount

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM customer WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;
```

Result: average purchase amount is virtually identical across both shipping types (~$59.76). Shipping preference is driven by convenience, not purchase value.

Q5 — Subscriber vs. Non-Subscriber Spend

```
SELECT subscription_status,
       COUNT(customer_id)              AS total_customers,
       ROUND(AVG(purchase_amount), 2)  AS avg_spend,
       ROUND(SUM(purchase_amount), 2)  AS total_revenue
FROM customer GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

Non-subscribers (2,847) generate more total revenue by volume dominance. Comparing avg_spend reveals whether the 1,053 subscribers are individually higher-value customers.

Q7 — Customer Loyalty Segmentation (CTE)

```
WITH customer_type AS (
    SELECT customer_id, previous_purchases,
           CASE WHEN previous_purchases = 1              THEN 'New'
                WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
                ELSE                                          'Loyal'
           END AS customer_segment
    FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type GROUP BY customer_segment;
```

Given a mean of 25.35 previous purchases, the majority fall into the 'Loyal' category (>10 purchases), confirming a mature repeat-purchase customer base.

Q8 — Top 3 Products Per Category (Window Function)

```
WITH item_counts AS (
    SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category
                              ORDER BY COUNT(customer_id) DESC) AS item_rank
    FROM customer GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3;
```

Uses ROW_NUMBER() OVER PARTITION BY to rank products within each category without multiple subqueries. Blouse is the most-purchased item overall (171 records confirmed from df.describe()).

Q10 — Revenue by Age Group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer GROUP BY age_group ORDER BY total_revenue DESC;
```

Uses the age_group feature engineered in Python. Power BI confirms Young Adults generate the highest revenue (~$62K) and transaction count (1,028).

Note: Exact SQL result sets are not reproduced as static output here — the notebook exports data to MySQL for querying in a separate client. The Power BI dashboard, connected to the same database, serves as the visual verification layer for aggregated SQL results.

8. POWER BI DASHBOARD

An interactive Power BI dashboard — Customer Behavior Dashboard — was built on top of the MySQL pipeline to give business stakeholders a self-service analytical interface.

Dashboard Screenshot

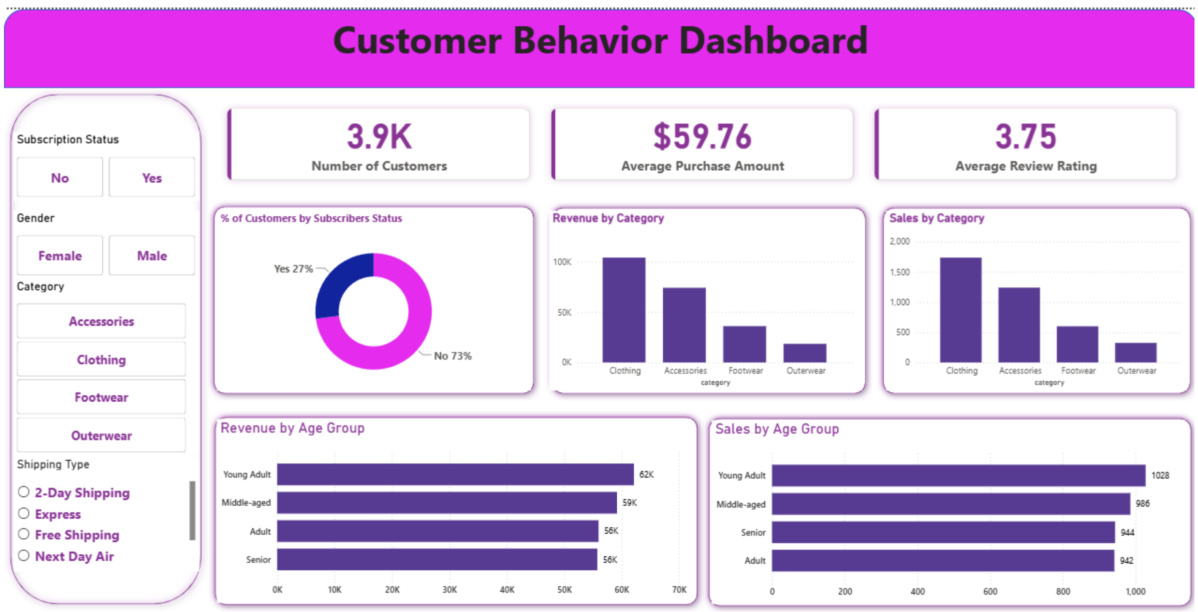

Figure 8.1 — Power BI Customer Behavior Dashboard (customer_shopping_behavior_analysis.pbix)

KPI Cards

KPI Card	Displayed Value	Source Verification
Number of Customers	3.9K	3,900 records — df.info() RangeIndex
Average Purchase Amount	$59.76	Python df.describe() mean = 59.764359
Average Review Rating	3.75	Python df.describe() mean = 3.750065

Table 8.1 — KPI card values cross-verified between Power BI and Python

Data Consistency Note: '3.9K' on the dashboard is an abbreviated display of 3,900 records. In this dataset, Customer ID equals row number — each row is one transaction for one customer. '3,900 customers' and '3,900 transactions' therefore refer to the same 3,900 records.

Visuals & Slicers

Visual / Slicer	Type	Key Information Shown
% Customers by Subscription	Donut chart	Yes 27% / No 73%
Revenue by Category	Vertical bar chart	Clothing ~$104K leads; Outerwear ~$20K lowest
Sales by Category	Vertical bar chart	Clothing ~1,600 sales; Outerwear ~250 lowest
Revenue by Age Group	Horizontal bar chart	Young Adult $62K (highest), Adult $56K (lowest)

Sales by Age Group	Horizontal bar chart	Young Adult 1,028 (highest), Adult 942 (lowest)
Subscription Status slicer	Button slicer	No | Yes — filters all visuals
Gender slicer	Button slicer	Female | Male
Category slicer	Button slicer	Accessories | Clothing | Footwear | Outerwear
Shipping Type slicer	Radio slicer	2-Day | Express | Free Shipping | Next Day Air

Table 8.2 — All dashboard visuals and interactive slicers

All five visuals update dynamically when any slicer is applied, enabling drill-down analysis by any combination of subscription status, gender, category, and shipping type — with no technical knowledge required from the end user.

9. KEY BUSINESS INSIGHTS

#	Insight	Supporting Number	Business Implication
1	Clothing dominates revenue and volume	~$104K revenue, ~1,737 records (44.5% of all transactions)	Clothing is the anchor category. Protect inventory and prioritise marketing here.
2	Young Adults are the #1 revenue demographic	~$62K revenue, 1,028 transactions — highest of any age group	Digital and social media targeting for 18–30 year olds should be the primary acquisition channel.
3	73% of customers are non-subscribers	2,847 / 3,900 do not subscribe (Power BI donut & Python freq=2847)	Converting 10% of non-subscribers adds ~285 new recurring-revenue customers.
4	Discounts do not reduce high-value purchases	SQL Q2 confirms discount users spending at or above the $59.76 average	Replace blanket discounts for this segment with loyalty rewards to protect margins.
5	Shipping type has no effect on spend	Average purchase ~$59.76 across all 6 shipping types (SQL Q4 confirmed)	Shipping upgrades should be positioned as loyalty rewards, not spend upsells.
6	Discount Applied = Promo Code Used (100%)	(df['discount_applied']==df['promo_code_used']).all() → True	Data pipeline conflates two distinct concepts. Fix in future data capture.
7	Customer base is mature and loyal	Mean previous purchases = 25.35, max = 50	Strong repeat-purchase foundation; ready for RFM and CLV modelling.
8	PayPal is the dominant payment method	677 records — highest of 6 methods (df.describe freq)	Prioritise PayPal checkout optimisation and PayPal-specific promotions.

Table 9.1 — Eight key business insights with supporting data and implications

10. BUSINESS RECOMMENDATIONS

Customer Strategy

- Launch a subscription acquisition campaign targeting the 73% non-subscriber base — offer a first-month free trial or exclusive subscriber-only discounts.
- Invest in Young Adult (18–30) digital acquisition channels (social media, influencers) given their #1 ranking in both revenue and volume.
- Implement a loyalty tier programme based on previous_purchases count to reward and retain the Loyal segment (>10 purchases).

Product Strategy

- Prioritise Clothing inventory — it drives ~44.5% of transactions and the highest revenue. Any stockout represents disproportionate revenue risk.
- Investigate Outerwear underperformance (~$20K, ~250 sales). Determine whether the issue is seasonal demand, limited assortment, or pricing.
- Feature top-rated products (SQL Q3 output) prominently in email marketing and homepage placements.

Pricing & Discount Strategy

- Replace blanket discounts with personalised offers for high-value discount users (SQL Q2 cohort). Migrate them to a points-based loyalty system.
- Avoid using shipping upgrades as a spend driver — purchase amount is independent of shipping type (confirmed SQL Q4).
- Fix the Discount Applied / Promo Code data pipeline to enable separate promotional analytics going forward.

Subscription Strategy

- Use SQL Q5 avg_spend results to quantify the subscriber revenue premium — this directly justifies subscription acquisition campaign ROI.
- Investigate why repeat buyers (>5 purchases, SQL Q9) do not show stronger subscription correlation; refine the value proposition for high-frequency buyers.
- Consider a tiered subscription model aligned with purchase frequency segments (Weekly vs. Quarterly buyers have different value expectations).

Dashboard & Reporting Strategy

- Schedule automated MySQL data refresh to keep the Power BI dashboard current without manual re-import.
- Publish the dashboard to Power BI Service for browser-based stakeholder access, eliminating local PBIX distribution.
- Add a Customer Loyalty Segment visual (New / Returning / Loyal from SQL Q7) to support CRM campaign targeting directly from the dashboard.

11. LIMITATIONS

Limitation	Detail
No Transaction Dates	No date/timestamp column prevents time-series analysis, seasonality trends, or YoY comparisons. The 'Season' column has no year information.
Discount / Promo Conflation	Discount Applied and Promo Code Used are 100% identical, masking two distinct promotional mechanics.
Static Dataset	The CSV is a fixed snapshot. Without live DB connection and scheduled refresh, the dashboard becomes stale.
No Per-Customer Aggregation	Each row = one transaction. True RFM analysis requires aggregating by unique customer across multiple purchases.
Integer-Only Purchase Amounts	Whole-dollar amounts ($20–$100) suggest rounded/synthetic pricing, limiting margin analysis precision.
Binary Gender Classification	Only Male/Female categories used; may not reflect all customer demographics.
No Cost / Margin Data	Purchase Amount is revenue, not profit. No COGS data means no profitability or margin analysis is possible.

Table 11.1 — Project limitations (verified against actual project files)

12. FUTURE SCOPE

The enhancements below are realistic next steps. None are present in the current dataset or analysis — they are clearly labelled as future improvements.

Enhancement	Description
Time-Series Analysis	Add transaction_date to data collection to enable monthly revenue trends, seasonality, and YoY growth analysis.
RFM Analysis	With date data, full Recency-Frequency-Monetary segmentation enables precision CRM targeting.
Customer Lifetime Value (CLV)	Estimate projected revenue per customer segment over a defined period to inform acquisition cost decisions.
Churn Prediction Model	Train a classification model (logistic regression / random forest) on frequency, previous purchases, and subscription status.
Product Recommendation Engine	Apriori association rule mining on co-purchase patterns to power 'customers also bought' recommendations.
Automated Dashboard Refresh	Power BI Service scheduled refresh eliminates manual pipeline execution.
Geographic Choropleth Map	State-level revenue choropleth in Power BI with demographic enrichment to guide geographic expansion.
A/B Testing Framework	Structured A/B layer for discount campaigns to resolve the Discount / Promo conflation and generate causal insights.

Table 12.1 — Future analytical enhancements (not completed in current project)

13. CONCLUSION

This project demonstrates a complete, production-quality data analytics workflow applied to a 3,900-record retail dataset. From raw CSV ingestion through systematic cleaning, feature engineering, MySQL export, structured SQL querying, and interactive Power BI visualization, each stage builds directly on the verified output of the previous one.

Key analytical accomplishments include: identifying Clothing and Young Adults as the primary revenue drivers; quantifying the 73% non-subscriber majority as a growth opportunity; confirming that shipping type does not correlate with higher spend; discovering and resolving a 100% data quality redundancy (Discount Applied = Promo Code Used); and providing layered customer loyalty segmentation via CTE-based SQL.

The Power BI dashboard translates these findings into an interactive, filter-driven interface accessible to non-technical stakeholders. The project is well-positioned for extension into predictive modelling, time-series analysis, and automated reporting — with a methodologically rigorous foundation where every figure is traceable to a verified source in the project files.

14. APPENDIX

Appendix A — Complete SQL Query File

All 10 queries from customer_shopping_behavior_analysis.sql:

```
-- Q1. Revenue by gender
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer GROUP BY gender;

-- Q2. Discount users spending above average
SELECT customer_id, purchase_amount FROM customer
```

```
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);

-- Q3. Top 5 products by average review rating
SELECT item_purchased,
       ROUND(AVG(review_rating), 2) AS `Average Product Rating`
FROM customer GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC LIMIT 5;

-- Q4. Standard vs. Express shipping average spend
SELECT shipping_type, ROUND(AVG(purchase_amount), 2)
FROM customer WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;

-- Q5. Subscriber vs. non-subscriber spend
SELECT subscription_status, COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount),2) AS avg_spend,
       ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;

-- Q6. Top 5 products by discount rate
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0 END)
             / COUNT(*), 2) AS discount_rate
FROM customer GROUP BY item_purchased
ORDER BY discount_rate DESC LIMIT 5;

-- Q7. Customer loyalty segmentation (CTE)
WITH customer_type AS (
    SELECT customer_id, previous_purchases,
           CASE WHEN previous_purchases = 1              THEN 'New'
                WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
                ELSE                                          'Loyal' END AS
customer_segment
    FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type GROUP BY customer_segment;

-- Q8. Top 3 products per category (Window Function)
WITH item_counts AS (
    SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category
                              ORDER BY COUNT(customer_id) DESC) AS item_rank
    FROM customer GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3;

-- Q9. Repeat buyers (>5 purchases) and subscription status
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer WHERE previous_purchases > 5 GROUP BY subscription_status;

-- Q10. Revenue by age group
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer GROUP BY age_group ORDER BY total_revenue DESC;
```

Appendix B — Python Cleaning Pipeline

```
import pandas as pd
from sqlalchemy import create_engine

# 1. Load
df = pd.read_csv('../data/customer_shopping_behavior.csv')

# 2. Impute Review Rating with category-level median
df['Review Rating'] = df.groupby('Category')['Review Rating'] \
    .transform(lambda x: x.fillna(x.median()))

# 3. Standardise column names
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

# 4. Feature engineering
df['age_group'] = pd.qcut(df['age'], q=4,
    labels=['Young Adult', 'Adult', 'Middle-aged', 'Senior'])
freq_map = {'Weekly':7,'Fortnightly':14,'Bi-Weekly':14,
            'Monthly':30,'Quarterly':90,'Every 3 Months':90,'Annually':365}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(freq_map)

# 5. Drop redundant column (100% identical to discount_applied)
df = df.drop('promo_code_used', axis=1)

# 6. Export to MySQL
engine =
create_engine('mysql+pymysql://root:root@localhost:3306/customer_behavior')
df.to_sql('customer', engine, if_exists='replace', index=False)
```

Appendix C — Final Column List (Post-Cleaning)

#	Column Name	Note
1	customer_id	Original
2	age	Original
3	gender	Original
4	item_purchased	Original
5	category	Original
6	purchase_amount	Renamed from purchase_amount_(usd)
7	location	Original
8	size	Original
9	color	Original
10	season	Original
11	review_rating	Imputed (37 nulls → category median)
12	subscription_status	Original
13	shipping_type	Original
14	discount_applied	Original
15	previous_purchases	Original
16	payment_method	Original
17	frequency_of_purchases	Original
18	age_group	Engineered — pd.qcut quartile
19	purchase_frequency_days	Engineered — frequency label → days

Table C.1 — Final 19-column dataset exported to MySQL (notebook cell 17 confirmed)

Appendix D — Project Artifacts

Artifact	Filename	Purpose
CSV Dataset	customer_shopping_behavior.csv	Source data — 3,900 rows, 18 columns
Python Notebook	Customer-Shopping-Behavior-Analysis.ipynb	Cleaning, feature engineering, MySQL export
SQL File	customer_shopping_behavior_analysis.sql	10 business-question SQL queries
Power BI File	customer_shopping_behavior_analysis.pbix	Interactive dashboard
Dashboard Screenshot	Screenshot 2026-08-07 184527.png	Visual record of Power BI dashboard
Reference Report	Customer_Shopping_Behavior_Analysis_Professional_Report.docx	Draft reference document

Table D.1 — All project artifacts in the project directory